

Xiyuan (Sophia) Shen, Meghan Howell, Sylvie Bon-Harper, Jeslyn Pratiknjo
Harlin Lee
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Predicting and Understanding Diabetes Risk

The Databetics: Xiyuan (Sophia) Shen, Meghan Howell, Sylvie Bon-Harper, Jeslyn Pratiknjo

1. Define the Problem

Every year, the Behavioral Risk Factor Surveillance System (BRFSS), which gathers telephone surveys on health, provides data to the CDC. To identify the characteristics that most closely correlate to diabetes, we examined the data that was gathered in 2015. Because we know people with diabetes, we found this topic to be interesting and relevant. Additionally, as the spread of the disease continues to grow, we would like to contribute to a solution. We proposed the question: How can we better identify individuals at high risk of diabetes before the condition becomes severe or irreversible by identifying the three most correlated features that are common in diabetic individuals? After considerable research and discussion, we developed our hypothesis, which is that blood pressure, physical activity, and body mass index (BMI) are the three most correlated features that are common in diabetic individuals.

2. Data Collection

In the data collection process, we gathered data from a variety of sources to develop our study. To learn more about the health-related telephone survey system used to collect health data from U.S. residents, we examined the article by the CDC, "Behavioral Risk Factor Surveillance System." It provides insight on how the data may be used to guide resource allocation, identify health patterns, and improve public health policies. We also reviewed the Honor Health article "Understanding excess weight and its role in type 2 diabetes" to understand more about the risk factors for type 2 diabetes. This ultimately helped us identify which features we believed to be most correlated with diabetes. Additionally, we used the "Diabetes Health Indicators Dataset" by Alex Teboul to explore the health data and code.

Within the dataset, there were 253,680 samples or survey respondents and 21 features. We chose to categorize the various features to better understand the data that was collected. We divided the features into five categories: chronic conditions and health indicators, preventative health and lifestyle, healthcare access, general and mental health, and demographics.

After examining the dataset, we discovered potential sampling bias and data collecting limitations. One potential limitation is that the dataset might not be representative of the whole population because it might disproportionately include people from particular socioeconomic groups, healthcare systems, or geographic areas. Moreover, the fact that the data is self-reported could introduce bias. Many features, like physical health and mental health, are susceptible to inaccuracies and underreporting. Another limitation is that there could be data imbalance. This implies that the study may be skewed by overrepresentation or underrepresentation of particular

features, such as the prevalence of diabetes and stroke. Lastly, the sampling methodology used for data collection was telephone surveys, indicating convenience sampling. This creates problems because it leaves out individuals who don't have access to telephones.

3. Data Preparation

We then moved into the process of preparing our data. The original data set includes a target variable named “Diabetes_012”, which classifies 0 as people without diabetes, 1 as people with prediabetes, and 2 as people with diabetes. We felt that it would be best if we dropped the prediabetes classifier so that we could focus solely on trends in people with and without diabetes. We carried this out by excluding every sample that was associated with prediabetes using the not equals (!=) operator. To keep everything uniform, we also used the *rename()* function to change the “Diabetes_012” column name to “Diabetes_02.”

Next, we noticed that there was an imbalance between individuals with and without diabetes in the dataset. To address this, we randomly selected a sample of 35,346 individuals without diabetes from the current dataset using the *sample()* function in pandas. We made this sample equal to a variable name “diabetes_0_new,” along with making the population of people with diabetes that we had equal to a variable named “diabetes_2,” and concatenated these two dataframes together using the *pd.concat()* function, naming the new dataframe “diabetes”. By observing equal amounts of people with and without diabetes from the data during our investigation, we could be sure that our observations would be balanced, thus improving the quality of our models.

Lastly, we addressed outliers in the data for key variables, including BMI, mental health score, physical health score, and age. To achieve this, we developed a custom *remove_outlier* function. This function accepts four inputs: a dataframe (table), a string (label) that will represent the column to process in the given table, and two values (Q1 and Q3) that represent the 25th and 75th percentiles of the variable, derived from its distribution. The function calculates the interquartile range (IQR) as $Q3 - Q1$ then uses this to compute the “lower_threshold” and “upper_threshold” variables, defining the range of typical values. Data points that fall below the “lower_threshold” or above the “upper_threshold” are considered outliers and are excluded from the dataframe. The function then returns the filtered dataframe, with the specified column processed and labeled according to the provided “label” parameter of the function. This function was repeated for all of our previously mentioned key variables, continuously updating our diabetes dataframe.

After all of this, we were finally prepared to start creating visuals of our data. All of the categorical variables had already been transformed into numeric form, so there was no need for further transformations there. Overall, these steps ensured that our dataset was clean and well-suited for analysis, which would therefore enhance the reliability of our predictive models.

4. Data Exploration

For our project proposal, we had two features we were particularly interested in. We used the *groupby* function to compare the difference between the mean BMI and the mean of physical activity of those who have diabetes and those who do not.

	BMI	PhysActivity
Diabetes_02		
0.0	27.190432	0.810348
2.0	30.435740	0.711959

This was a helpful statistic because it showed us that there was a difference in BMIs and physical activity between people with diabetes and people without. Because it's a mean, it does not matter that the data set is unbalanced, because it's taking the average.

In continued data exploration, we counted the number of diabetics and non-diabetics in the data set. We then removed the outliers from “BMI”, “MentHlth” score, “PhysHlth” score, and “Age” using a custom function. We also struggled with the categorical data aspect of this data set. Because many of the features were classified by either 0 or 1, or in the target feature case, 0 or 2, it was challenging to do regression or other tools we learned about in class. We also calculated the correlation between each of our features and our target. Below are the results:

```
diabetes['Diabetes_02'].corr(diabetes['PhysActivity'])
```

```
np.float64(-0.11556597906014833)
```

```
diabetes['Diabetes_02'].corr(diabetes['BMI'])
```

```
np.float64(0.3024854968489143)
```

```
diabetes['Diabetes_02'].corr(diabetes['HighBP'])
```

```
np.float64(0.3702740020564629)
```

5. Model Building

We decided to use a Decision Tree Classifier from sklearn to construct our model. Since we will use this classifier multiple times in our codes, we created a function called *decision_tree* with selected features, max depth, and the table diabetes as the arguments. The model takes 80% of the dataset as training data and 20% as testing data. The model outputs a decision tree and the accuracy score on train and accuracy score on test.

The reason we chose to use decision trees is because our goal is to 1) determine the three most correlated features to diabetes and 2) be able to predict whether a new patient has diabetes or not based on known measures of predictor variables. A decision tree is best at performing classification, outputting a visual that ranks nodes based on their importance, and determining what class a sample might be, given its features. Consequently, we will be able to select the top three nodes of the tree as our top three features.

Furthermore, we are working with different types of data. Most of the features we are working with are categorical, such as “Diabetes_02,” but a few are numerical, like “BMI.” A decision tree model can work with both of these types of data efficiently. On the contrary, a linear regression model wouldn’t be helpful because it is used primarily with numerical data, and thus it wouldn’t capture the complexity of our problem. k-NN wouldn’t be useful either, as it doesn’t tell us which features are important in classification.

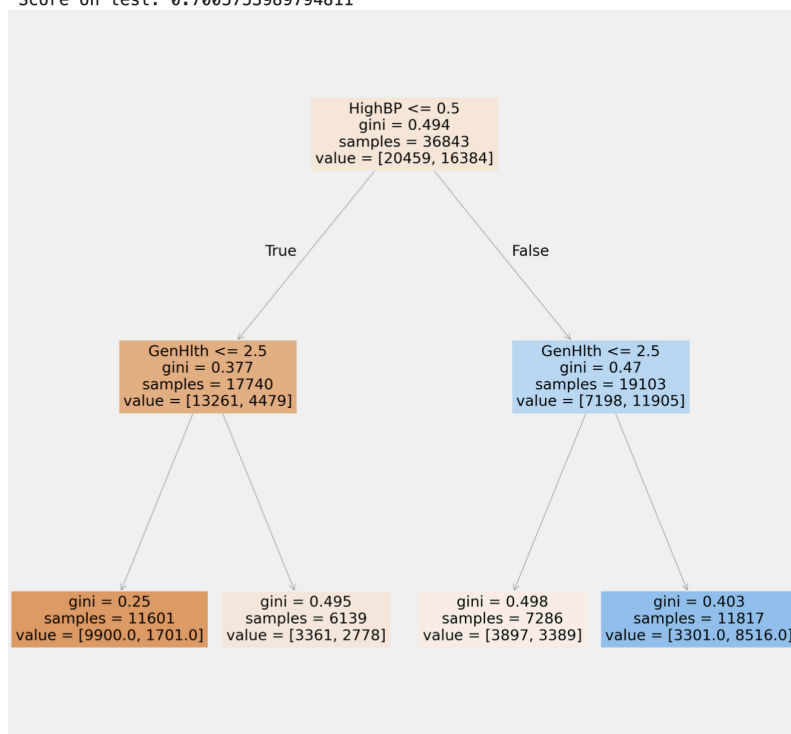
Methodology

i) Decision tree without feature selection:

Once we built our model, we first inputted all the 21 features and experimented with depths 2, 3, and 4.

Max depth = 2

```
0.0    6305
2.0    2906
Name: count, dtype: int64
Score on train: 0.6968487908150802
Score on test: 0.7005753989794811
```



Max depth = 3

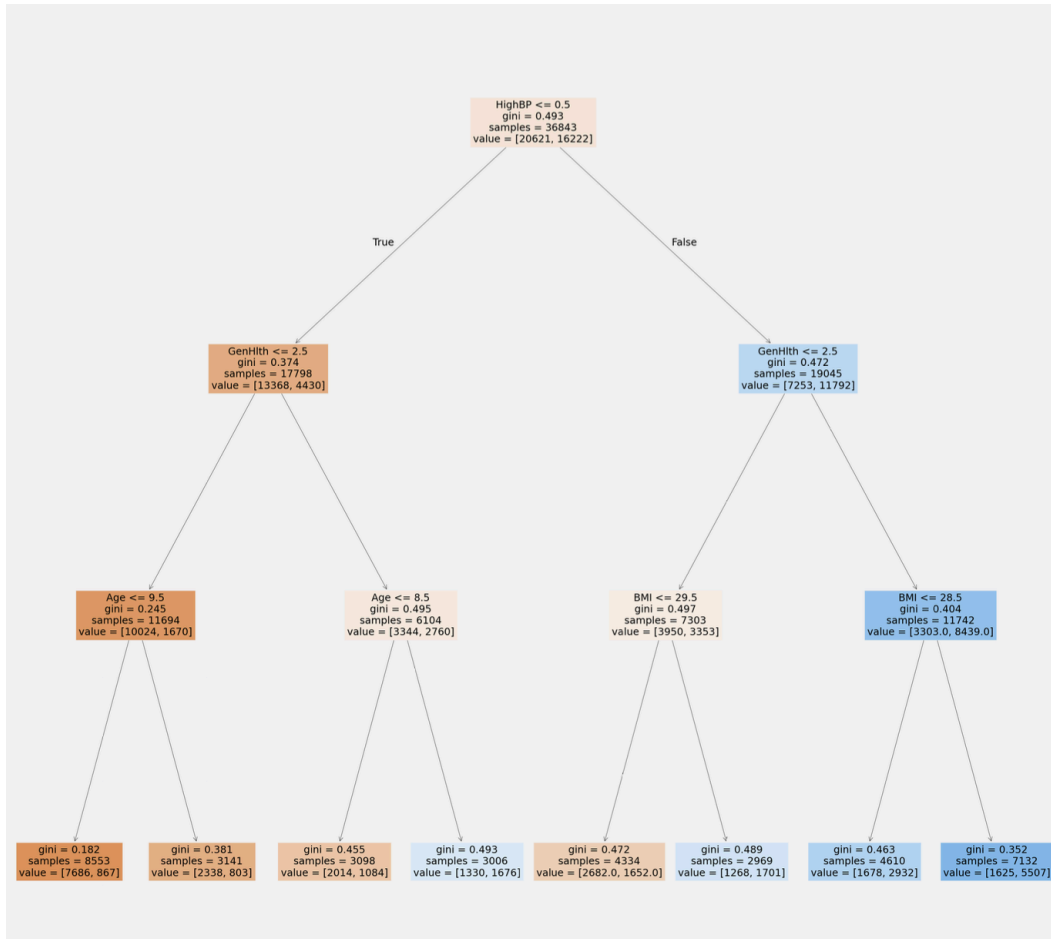
0.0 4727

2.0 4484

Name: count, dtype: int64

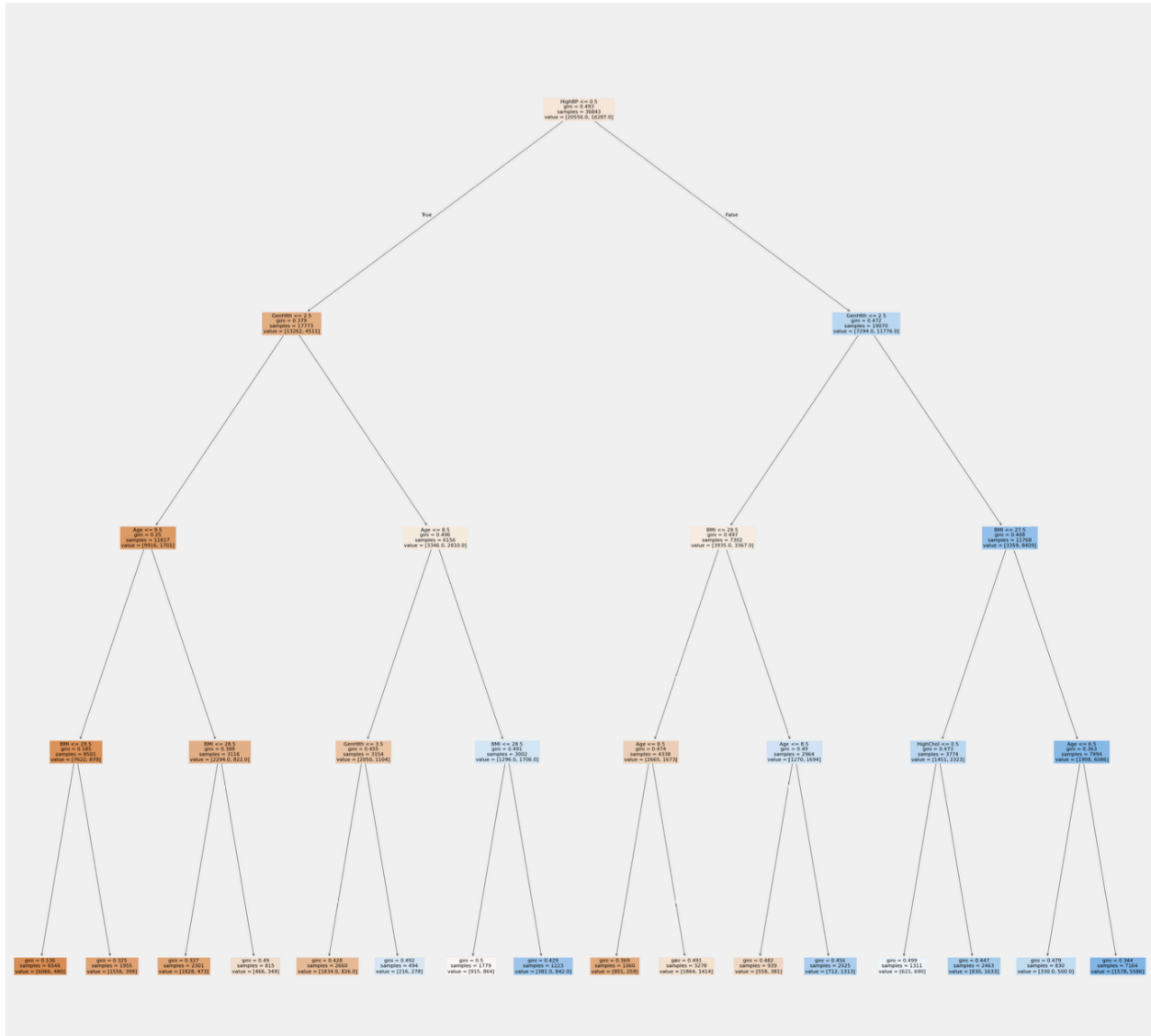
Score on train: 0.7202453654696958

Score on test: 0.7107805884268809



Max depth = 4:

```
0.0    5298
2.0    3913
Name: count, dtype: int64
Score on train: 0.725510951876883
Score on test: 0.7322766257735317
```



ii) Decision tree with feature selection:

Then, to reduce the number of features the classifier has to process at one time, we performed feature selection. This will theoretically make the model less complex, decrease the probability of overfitting, and, overall, optimize our model. We created an algorithm where the computer would randomly select 5 features from the 21 features, input those into the decision tree model with a depth of 3, repeat 1,000 times, and output the 5 features that allowed for the

highest test and train scores. This algorithm was run three times, so in the end, we had three combinations of 5 features. We then re-evaluated each combination using *decision_tree* to verify whether the 5 features in the combination had relatively high accuracy scores. Below are the three combinations of feature selection along with its corresponding decision tree and accuracy scores.

Combination 1:

Best Features: ["Diabetes_02", "Fruits", "GenHlth", "Age", "BMI", "NoDochbcCost"]

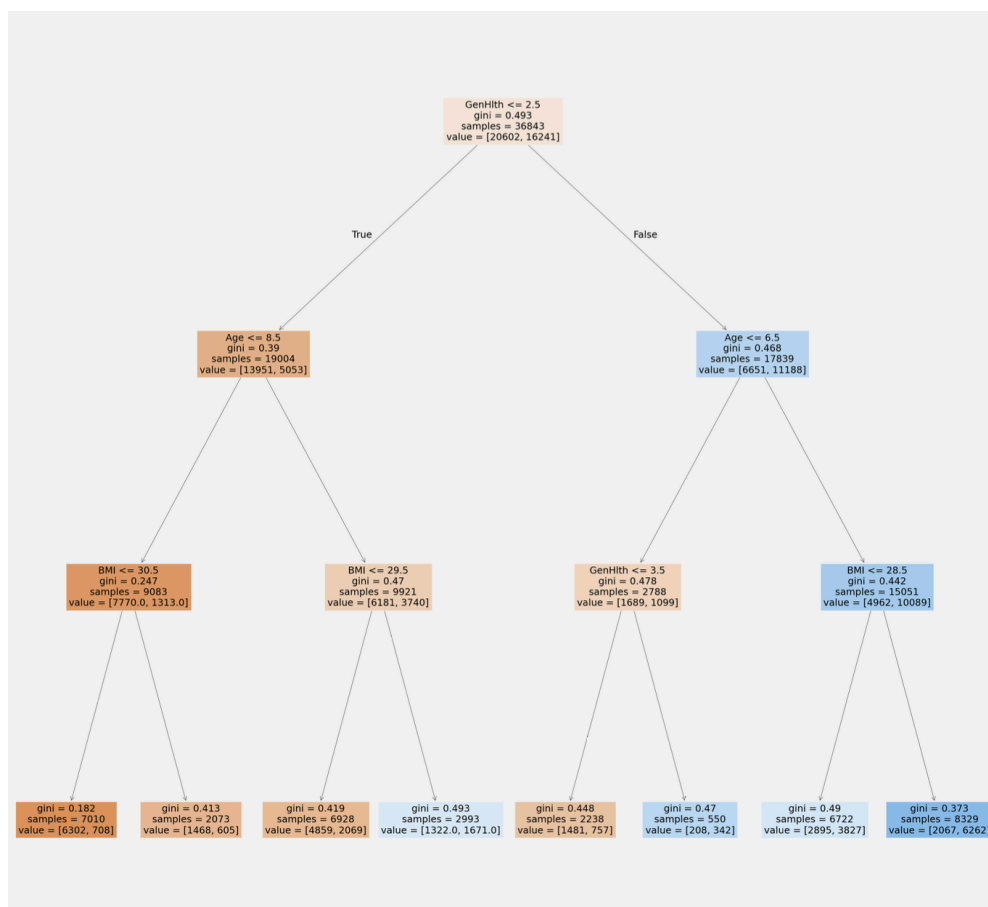
2.0 4736

0.0 4475

Name: count, dtype: int64

Score on train: 0.7114512933257335

Score on test: 0.7130604711757681



Combination 2:

Best Features: ["Diabetes_02", "HighBP", "BMI", "Age", "GenHlth", "Income"]

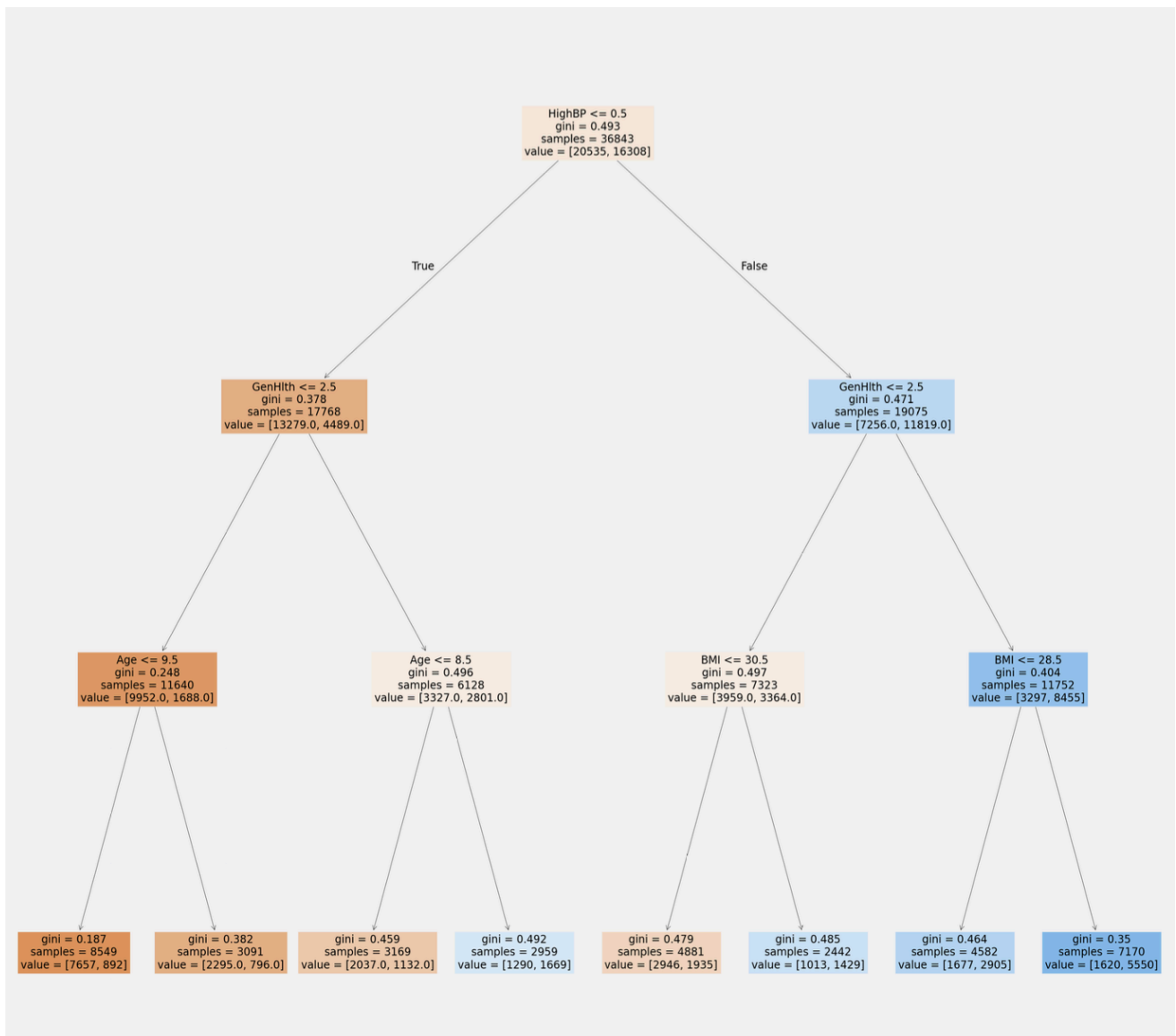
0.0 4860

2.0 4351

Name: count, dtype: int64

Score on train: 0.7189425399668865

Score on test: 0.7184887634350233



Combination 3:

Best Features: ["Diabetes_02", "Age", "GenHlth", "HighBP", "Income", "BMI"]

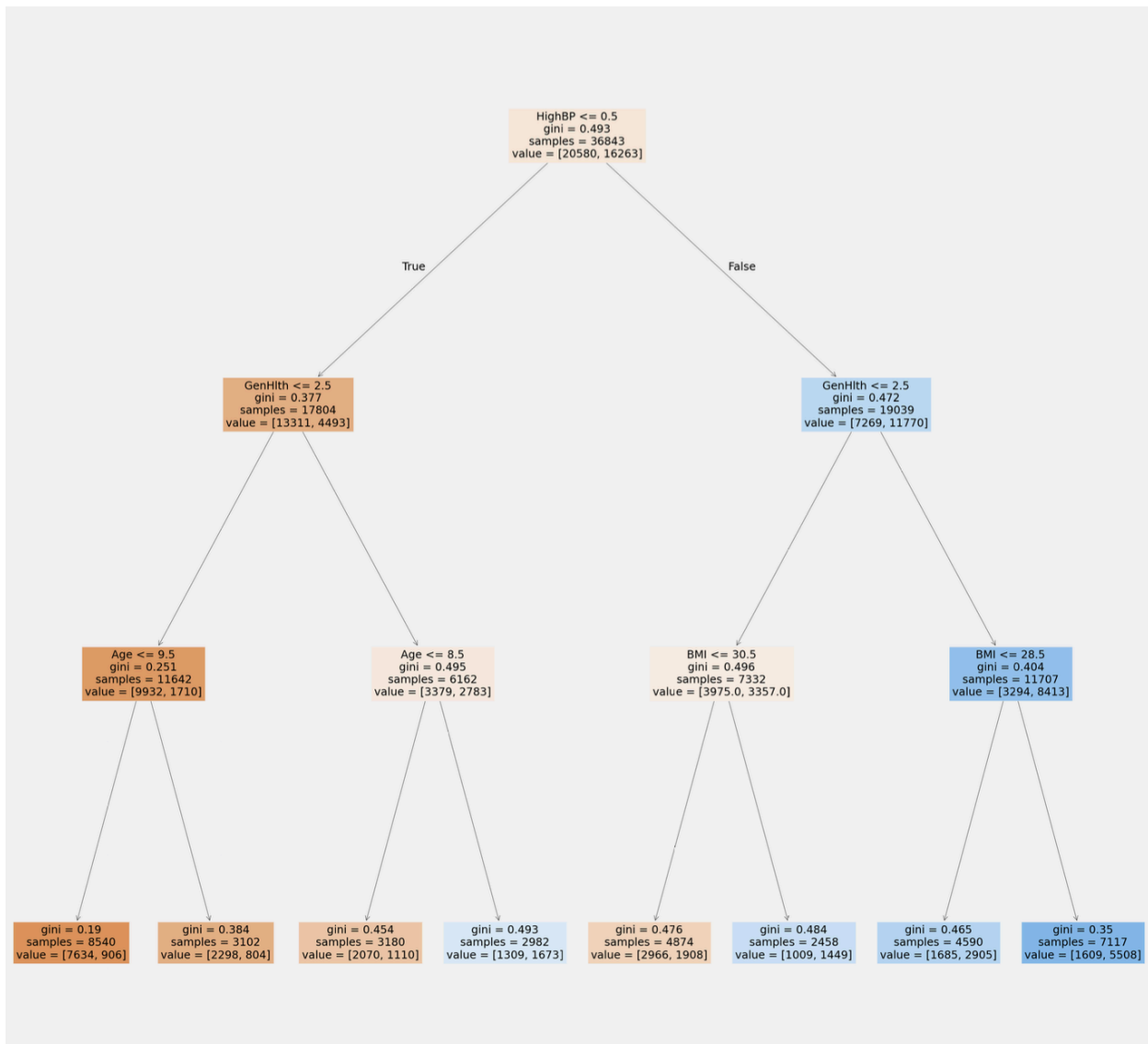
0.0 4854

2.0 4357

Name: count, dtype: int64

Score on train: 0.7193496729365144

Score on test: 0.7168602757572468



6. Model Evaluation

i) Decision tree without feature selection:

As we increase the depth of the model, the accuracy score on both test and train increases. When max depth is 3, the train score is 0.72 and the test score is 0.71. As the discrepancy between train score and test score is relatively small, the model we used did not overfit and can be generalized to new test sets.

ii) Decision tree with feature selection:

While we experimented with feature selection, we chose to use depths of 3. This is because, according to what was displayed in the trees without feature selection, if the maximum depth is larger than 3, the tree becomes more complicated to read. Furthermore, since a depth of 3 does not lead to overfitting and we are only choosing the three top features, a depth of 3 is sufficient to help us make conclusions.

Given that the train and test scores are similar, the test accuracy scores of the trees with feature selection are about the same as those without feature selection, which is around 0.715. This represents that our model can classify well without the distraction of irrelevant data and that the elimination of feature selection did not majorly degrade the performance of our model.

iii) Comparison between decision tree and random forest

According to research and suggestions by TAs, we decided to construct a random forest model, which trains the dataset through multiple decision trees with bootstrapped subsets of features to output one result of classification. Yet the drawback of this model is that it doesn't show the process of how it classifies nor displays the trees, thus we do not know what the top three nodes are. Instead of using random forest as our original model, we decided to use it as a standard to compare our model. Supposedly, the accuracy score of random forest should be equal to or higher than the score of decision trees because it decreases bias and removes any correlation between features. Our random forest model, with $n_estimators = 1000$ and $max_depth = 3$, outputs a test accuracy score of 0.73 which is close to the test accuracy scores of our tree model. This again represents that our model performs well even without feature selection.

iv) Relating to our hypothesis

In relation to our hypothesis, a majority of our decision trees conclude that “HighBP” and “GenHlth” are indicative features of our target variable. Furthermore, “BMI” appeared in all combinations of selected features and the average was higher for people with diabetes (diabetes = 2) than for people without (diabetes = 0) as shown in the summary table in Step 4: Data

Exploration. This corroborates our hypothesis that blood pressure and BMI are indicative variables in predicting whether a person has diabetes or not.

However, our model did not show that “PhysActivity” is a strong indicator of diabetes, which suggests that our hypothesis might be inadequate for choosing this index as one of the top three important features. The weak linear correlation between “Diabetes_02” and “PhysActivity,” which is around -0.12, also corroborates the claim. Consequently, based on our model, “PhysActivity” is not a top three feature of predicting diabetes, instead, “GenHlth” might be more predictive.

7. Model Deployment

We envisioned this model being used to identify people at risk for developing diabetes. This model could not be released for the general public to use without qualifications. The model cannot, and should not be used to, diagnose patients with diabetes. The model also does not prove causation. We are not saying that the strongest correlated features cause diabetes, only that there is a link between them. This model should be used by healthcare providers who can do testing if the model provides evidence suggesting a patient may be at risk for diabetes.

There is a risk of over reliance on the model. If the model does not say a patient is at risk of diabetes, they may still have or be at risk of the disease. Our accuracy rate is not 100%, so the model cannot be blindly followed. Medical providers need to use their other training to diagnose or prescribe preventative measures to stop this disease.

References

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